IOT DAY · RV UNIVERSITY

A Low-Cost Multi-Sensor IoT Rig for Process-Aware Lab-Dip Shade Assessment

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1 Aim

To build a **low-cost IoT rig** that records every colour reading of a textile lab dip **together with the dye-bath and room conditions** that produced it, tagged with the sample ID, so that a failed dip can be **traced to a likely cause** instead of guessed at.

2 Problem Statement

- A **lab dip** is a small swatch dyed to the buyer's shade before bulk dyeing.
- When a dip is rejected, the recipe is blamed, yet **bath temperature**, **lighting** and **humidity** also shift the shade.
- Labs rarely record these beside the colour reading, so a rejected dip has **no cause attached**. Each extra round wastes dye, water and a day.
- Spectrophotometers are costly and do not log the process.

3 Components Used

HARDWARE

ESP32	reads every sensor each 3 s, colour → CIE Lab, ΔE, MQTT
TCS34725	fabric colour sensor with its own LED (I²C)
ESP32-CAM	swatch photo for colour and levelness, flash LED
DS18B20	waterproof dye-bath temperature probe
DHT11	room temperature and humidity
BH1750	ambient light level (lux)
SSD1306 OLED	live readings and bath alert on the rig
Measurement box	21 × 16 × 8.5 cm, white-lined, light-sealed

SOFTWARE

MQTT broker	one topic each for readings, commands, camera results
Android app	LabDip Monitor (Kotlin, Compose): live ΔE, camera card, alerts, trends, rig controls
Python	logger, live comparison, OpenCV camera analysis, statistics

4 Working Principle

Sense colour, bath, room, light	**ESP32** Lab, ΔE, bath alert	**MQTT** JSON every 3 s

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App ΔE, PASS / FAIL, commands	**Laptop** log, compare, analyse	**ESP32-CAM** photo → Lab, grid

1. **Colour:** TCS34725 counts ÷ clear channel, scaled to 0–255, corrected by calibration gains, then sRGB → XYZ (D65) → CIE L*a*b*.
2. **Target:** the reference swatch is captured from the app. Every later swatch gets ΔE and PASS / FAIL against it.
3. **Process:** bath temperature (alert beyond ±2 °C of set point), room temperature, humidity and light go in the same record.
4. **Camera:** each photo is corrected pixel by pixel with a white-paper photo, the fabric patch is found automatically and a 3 × 3 grid checks dyeing levelness. Results show on the app's camera card.
5. **Analysis:** correlation, OLS with ANOVA and cross-validated regression rank the most likely cause of a colour deviation.
6. **Safety:** every command is range-checked. A missing sensor sends nothing, never a fake value.

5 Measurement Box & Calibration

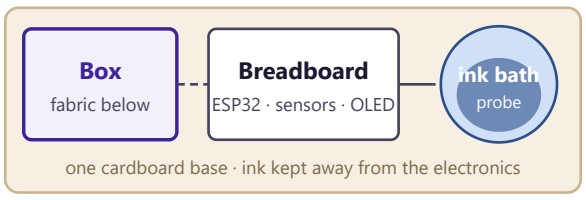
fabric swatch (open side down)

- The camera's own flash bounces off the white ceiling: soft light, no glare, no extra LEDs.
- The TCS sits in its own opaque box a few mm above the fabric.
- A foam rim seals out room light. The room sensors stay outside on the breadboard.

CALIBRATED ON PLAIN WHITE PAPER

TCS34725	gain 1×, 154 ms. Per-channel gains R 1.0, G 1.585, B 2.708 turn white paper from R130 G82 B48 into neutral grey.
ESP32-CAM	white balance and exposure locked (exposure 150, no clipping). A white-paper photo corrects every later photo pixel by pixel.
Re-check	take a new white photo whenever the box, camera or lighting changes.

6 Live Demo



A glass of water with ink stands in for the dye bath; dyed fabric swatches go under the box.

1. Reference fabric under the box, tap **Capture** in the app: ΔE ≈ 0.
2. Swap in another shade: ΔE rises, PASS / FAIL updates on OLED and phone.
3. Camera photo: Lab value and levelness grid appear on the app.
4. Warm the probe by hand out of the ink: bath alert beyond ±2 °C.

7 Results

Real rig, white-paper calibration

White paper on the TCS34725 now reads neutral (a −0.14, b 0.22). The camera's green cast is removed by the white reference: a yellow swatch reads **L 78.4, a −9.7, b 48.3**. Both sensors agree on its hue: **98° vs 101°**.

ANALYSIS ON SYNTHETIC DATA 60 simulated lab dips, 5,464 rows

0.806	0.727	0.529
OLS R^2	Linear reg. CV R^2	Random forest CV R^2



Which condition drives ΔE (permutation importance). The simpler linear model beat the random forest. This checks the method, not real dyeing.

Rig ΔE is CIE76; camera and analysis use CIEDE2000. The two are not compared directly.

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